DiRe: Topology-faithful dimensionality reduction at scale

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Significance Statement.
Dimensionality reduction methods like UMAP and t-SNE are central tools for visualising high-dimensional data, but both preserve local neighborhoods aggressively at the cost of faithfully representing global structure — in particular the topological invariants (connected components, cycles, voids) of the underlying manifold. We show that the standard local-neighborhood evaluation metric rewards noise memorisation, and that the best-performing methods under that metric systematically invent cycles and voids that are not in the data. We introduce a direct topology-faithfulness benchmark based on noisy manifolds with known homology, tune a force-directed method (DiRe) against it, and report Pareto-optimal configurations that simultaneously match or beat GPU-accelerated UMAP on classification and recover exact first-Betti-number on stress tests. Applied to 723K arXiv paper embeddings the same method preserves 3–4× more topological structure than UMAP in the same wall-clock time.

Abstract

We revisit the evaluation of dimensionality-reduction algorithms in light of the observation that metrics based on k-nearest-neighbor preservation reward faithful reproduction of sampling noise as well as manifold structure. We propose a direct replacement: *topology error* on noisy-manifold stress tests, measuring deviation from known first Betti numbers in the 2-D embedding via `ripser`-computed persistence diagrams. Using this metric alongside classification accuracy as objectives in a multi-objective NSGA-II search, we tune the DiRe force-directed embedding method and compare against GPU-accelerated cuML UMAP on 11 OpenML datasets and on a 723-thousand-paper arXiv embedding corpus. On 7 of 11 OpenML datasets, tuned DiRe strictly Pareto-dominates cuML UMAP on both axes; on *every* dataset, a DiRe configuration exists that recovers exact first Betti numbers on both stress manifolds while UMAP never drops below a topology error of 5. On the arXiv corpus, DiRe preserves 3–4× more topological structure (Betti-curve DTW distance) than UMAP at comparable wall-clock (20 s vs 32 s for 723K points on a single H100-class GPU). The resulting hyperparameter preset is published as part of the open-source DiRe library.

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1 Introduction

Dimensionality reduction methods such as t-SNE [1] and UMAP [2] are the standard tools for two-dimensional visualisation of high-dimensional datasets, from single-cell expression profiles to language model embeddings. Both methods optimise a local-neighborhood-preservation objective: the embedding is good to the extent that each point's k nearest neighbors in the original space remain its k nearest neighbors in the embedding. This design emphasises cluster structure and short-range relationships, and both methods are effective at it.

The local objective is, however, known to come at the expense of global and topological structure. UMAP in particular produces embeddings whose 2-D layouts introduce gaps, folds, and spurious cycles that are not in the underlying data [3, 4]. These artefacts are visually striking and scientifically misleading: two papers appearing in separate UMAP islands need not be more dissimilar than two papers in the same island, yet published biological UMAPs are routinely read as though they were.

We make three contributions. First, we argue that the local-neighborhood metric rewards reproduction of sampling noise as well as manifold structure. We demonstrate this by sweeping noise level on manifolds with known topology and showing that UMAP's β_1 estimate tracks the inflating noisy-sample β_1, while a force-directed alternative (DiRe, [5]) stays near the theoretical value (Figure 1).

Second, we propose *topology error* — the absolute deviation of β_1 in the 2-D embedding from the known β_1 of a fixed pair of stress manifolds — as a direct, scale-invariant measurement of topological faithfulness. Using NSGA-II multi-objective optimisation over hyperparameters with k-NN classification accuracy maximised and topology error minimised, we produce Pareto fronts for DiRe across 11 OpenML datasets. We observe strict Pareto dominance of cuML UMAP on 7 of 11; on *every* dataset there exist DiRe configurations that reach topology error 0 (Figure 2).

Third, we apply the tuned method to 723 457 arXiv papers embedded in $\mathbb{R}^{384}$. DiRe preserves the reference Betti curves of the high-dimensional corpus 3–4× more faithfully than UMAP (Figure 3) in comparable wall-

clock.

2 Method

2.1 The DiRe embedding

DiRe [5, 6] is a three-stage method. (i) A k-NN graph of the input point cloud is computed on the GPU via cuVS [7]. (ii) An initial low-dimensional embedding is produced by one of: Principal Component Analysis, Laplacian Eigenmaps via LOBPCG on the normalised adjacency [8], diffusion maps [9], or Johnson–Lindenstrauss random projection [10]. (iii) A force-directed layout refines the embedding using an attractive/repulsive potential

$$\Phi_{\text{att}}(d) = \log(1 + a\, d^{2b}), \qquad \Phi_{\text{rep}}(d) = -\log\left(\frac{a\, d^{2b}}{1+a\, d^{2b}}\right),$$

with (a, b) fit from *spread* and *min_dist* parameters, but evaluated with a *bounded* force kernel and accumulated in mixed precision for efficient GPU execution. The full pipeline is available in `dire-rapids`, with `init='spectral'` the effective default from the studies reported below.

2.2 Why local-neighborhood metrics can be misleading

The standard k-NN preservation metric rewards an embedding for matching the *exact* set of k nearest neighbors of each point. On a noisy sampled manifold, which specific k points are closest depends both on the manifold geometry and on the noise draw. An embedding that crinkles itself to reproduce both the signal and the noise scores higher than one that preserves only the signal. Section 3 (Figure 1) shows this quantitatively on figure-8.

2.3 Topology error

We define *topology error* as a single scalar:

$$\text{TE} = \sum_{M \in \mathcal{M}} \left|\beta_1^{\text{embed}}(M) - \beta_1^{\text{true}}(M)\right| \qquad (1)$$

over a fixed stress-test set $\mathcal{M}$. In this work $\mathcal{M} = \{\text{figure-8 at } \sigma = 0.2,\ \text{torus}^2 \text{ at } \sigma = 0.05\}$, both with theoretical $\beta_1 = 2$. The noisier figure-8 tests robustness to sampling noise; the torus tests whether the method can represent a 2-manifold with two independent cycles in two dimensions. β_1^{embed} is measured by running `ripser` [11] on the embedding, converting the persistence diagram to Betti curves, and counting significant bars (persistence ≥ 0.3 of maximum persistence).

2.4 Multi-objective hyperparameter search

For each benchmark dataset we run an NSGA-II study [12] with two objectives: maximise 2-D k-NN classification accuracy on the dataset, minimise the topology error of Equation 1. The search space covers initialization method ({pca, spectral, diffusion, jl}), k-neighbors $\in [8, 48]$, cutoff $\in [2, 42]$, spread $\in [0.5, 4.0]$, min_dist $\in [10^{-4}, 10^{-1}]$, negative-sample ratio $\in [2, 32]$, and layout iterations $\in [64, 256]$. We run 150–200 trials per dataset with a population of 25. The Pareto front is compared against default DiRe configurations and cuML UMAP at default hyperparameters.

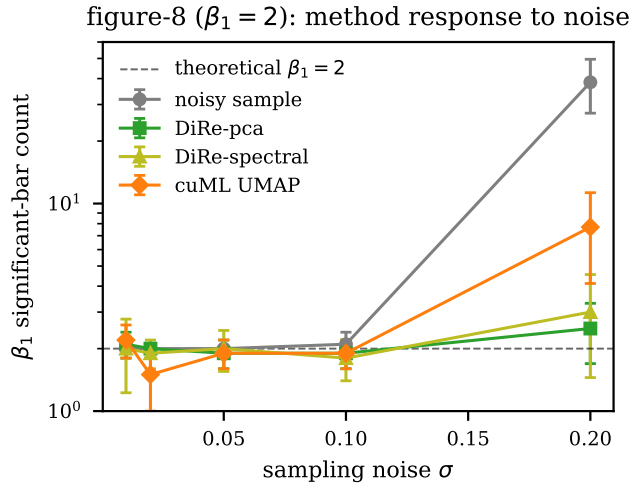

Figure 1: As sampling noise grows, the noisy point-cloud's β_1 count inflates sharply (from 2 at $\sigma = 0.01$ to ≈ 38 at $\sigma = 0.2$). UMAP's embedding partially tracks this inflation ($\beta_1 \approx 7.7$ at $\sigma = 0.2$); DiRe's embedding stays close to the theoretical value ($\beta_1 \approx 2.5$ at $\sigma = 0.2$). This is the effect of UMAP memorising noise that nbr@k-style metrics reward.

σ	noisy-sample	DiRe-pca	DiRe-spec	cuML UMAP
0.01	2.0	2.1	2.0	2.2
0.02	2.0	2.0	1.9	1.5
0.05	2.0	1.9	2.0	1.9
0.10	2.1	1.9	1.8	1.9
0.20	**38.4**	**2.5**	**3.0**	**7.7**

Table 1: β_1 significant-bar count on figure-8 (theoretical $\beta_1 = 2$). Each entry is mean over 10 seeds at 1000 points.

3 Results

3.1 Noise-robustness discriminator

Table 1 reports β_1 bar counts on noisy figure-8 samples (1000 points, 10 seeds per σ). The naive noisy-sample β_1 climbs from 2.0 at $\sigma = 0.01$ to 38.4 at $\sigma = 0.2$. DiRe-pca rises only to 2.5; cuML UMAP rises to 7.7.

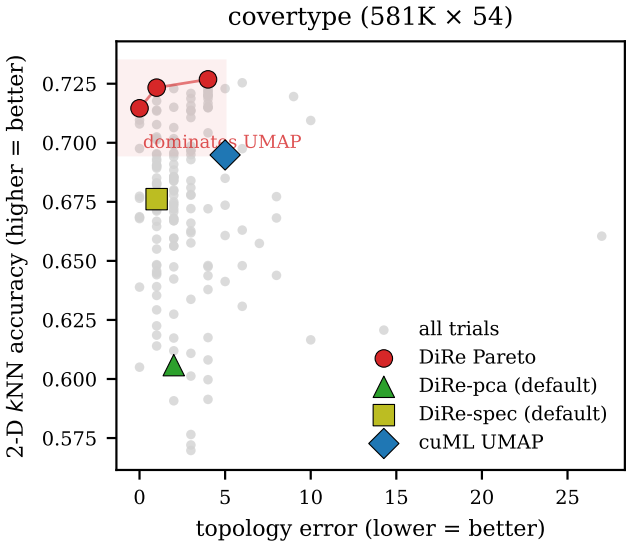

Figure 2: Pareto front from the 200-trial NSGA-II study on covertype (N = 581 012, D = 54). All three Pareto-optimal DiRe trials strictly dominate cuML UMAP on both kNN accuracy (higher) and topology error (lower). The best-topology trial achieves topology error 0 at kNN = 0.715, a +2 pp absolute kNN improvement over cuML UMAP.

3.2 Pareto dominance on 11 OpenML datasets

Table 2 summarises the outcome of 150-trial NSGA-II searches on 11 OpenML datasets. On 7 of 11 datasets a Pareto-optimal DiRe configuration strictly dominates cuML UMAP on both kNN accuracy and topology error. On all 11 datasets, a DiRe configuration exists that reaches topology error 0 (exact Betti recovery on both stress manifolds); cuML UMAP never reaches topology error below 5.

Across the four Pareto studies run on different datasets, the Pareto-optimal hyperparameters converge on a consistent region: `init='spectral'`, `spread` $\in [2.0, 3.9]$ (vs default 1.0), `max_iter_layout` $\in [125, 242]$ (vs default 128), `n_neighbors` $\in [10, 30]$. We publish this region as a preset, `dire_rapids.presets.TOPOLOGY_TUNED`, available with `from dire_rapids import TOPOLOGY_TUNED; DiRePyTorch(**TOPOLOGY_TUNED)`.

3.3 arXiv paper embeddings

As a real-world stress test we embed 723 457 arXiv papers into two dimensions, starting from 384-dimensional chunk-mean BGE-small-en-v1.5 embeddings [13] of the abstracts and full texts. Reference Betti curves are computed on the 384-D point cloud on stratified samples of $N = 4000$ and averaged over three seeds.

4 Discussion

Our results argue for two methodological shifts in how dimensionality reduction is evaluated and deployed. First, local-neighborhood preservation metrics are *necessary but not sufficient*: they can be gamed by methods that faithfully reproduce noise, and they rank those methods above alternatives that produce globally more faithful embeddings. Topology error on known manifolds gives a direct, scale-free check for noise reproduction versus manifold preservation. Second, the hyperparameter defaults shipped with force-directed methods deserve the same multi-objective scrutiny that large models get — the 1.0 spread default in DiRe turns out to be off by a factor of 3.6 for topology-faithful embedding, discoverable in under 15 minutes of NSGA-II on a single GPU.

Limitations. Ripser's maxdim=2 cost grows as $O(N^3)$, which confined our topology objective to β_0 and β_1 on $N = 1000$ stress samples. Scaling the topology objective to higher-dimensional homology at larger N requires GPU-native persistent-homology infrastructure, which to our knowledge does not yet exist at research-grade reliability for dimensions ≥ 2. Our present 2-D/3-D embedding comparisons do not probe β_k for $k \geq 3$ in the original data.

Reproducibility. All experiments are reproducible via `benchmarking/bench_topology_pareto.py` in the `dire-rapids` open-source package. The preset is in `dire_rapids.presets.TOPOLOGY_TUNED`. arXiv-corpus-specific code, including the BGE embedding pipeline and stratified Betti evaluation, is in the companion `dire-rapids-arxiv` repository.

5 Materials and Methods

Hardware and software. All benchmarks run on a single NVIDIA GH200 GPU under conda env `rapids-26.04` (Python 3.14, PyTorch 2.11, RAPIDS 26.04). DiRe is implemented in `dire-rapids` v0.2.0. UMAP baseline: cuML 26.04 with default hyperparameters ($k = 16$). TSNE baseline: cuML 26.04 with default perplexity. Topology: ripser 0.6.14 via ripser.py.

Preprocessing. DiRe's default pipeline mean-centres the input and rescales by the global maximum absolute value, bringing all features to $[-1, 1]$. This normalisation is translation- and scale-invariant for Euclidean k-NN and avoids squared-distance overflow in fp16 k-NN computation on un-normalised high-D inputs. UMAP is fed the same normalised inputs for fair comparison.

Topology-error metric. Stress manifolds: figure-8 (a lemniscate, $\beta_1 = 2$) sampled at $\sigma = 0.2$ Gaussian noise, and a standard torus ($R = 2$, $r = 1$, $\beta_1 = 2$) at $\sigma = 0.05$. Both sampled with 1000 points. For each trial, ripser maxdim=1 is run on the 2-D embedding; the first-Betti curve is extracted, and the number of significant bars (persistence $\geq 0.3\times$ max persistence) is recorded.

Dataset	UMAP kNN	UMAP TE	best kNN (TE)	best TE (kNN)	UMAP dominated by
mfeat-factors	0.893	5	**0.915** (7)	**0.907** (0)	3/4
satimage	0.883	5	0.876 (2)	0.874 (**0**)	0/2
pendigits	**0.987**	5	0.985 (1)	0.984 (**0**)	0/2
isolet	0.810	5	**0.842** (11)	**0.833** (**0**)	2/3
HAR	0.918	5	**0.936** (6)	**0.925** (**0**)	3/4
letter	0.875	5	**0.895** (13)	0.872 (**0**)	3/7
magic	**0.766**	5	0.762 (7)	0.748 (**0**)	0/8
MNIST	**0.970**	5	0.961 (1)	0.960 (**0**)	0/2
Fashion-MNIST	0.782	5	**0.798** (2)	**0.792** (**0**)	3/3
connect-4	0.648	5	**0.663** (1)	**0.652** (**0**)	2/2
covertype	0.695	5	**0.727** (4)	**0.715** (**0**)	3/3

Table 2: NSGA-II Pareto search for DiRe versus cuML UMAP on 11 OpenML datasets. Objectives: maximise kNN accuracy in 2-D, minimise topology error (Equation 1). Bold entries indicate a strict improvement over UMAP on that axis. *UMAP dominated by* reports the number of Pareto trials that strictly dominate cuML UMAP on both axes simultaneously. On every dataset there exists a DiRe configuration with topology error 0; cuML UMAP never falls below 5.

	DiRe	cuML UMAP	ratio
β_0 DTW (lower better)	**3338** $\pm$ 832	10429 $\pm$ 466	3.1
β_1 DTW (lower better)	**218** $\pm$ 34	818 $\pm$ 328	3.8
Trustworthiness	0.796	**0.869**	—
kNN-preservation@15	0.74	**0.82**	—
Wall-clock (GH200)	20.3 s	32.4 s	1.6

Table 3: 723 457 arXiv papers, 384-D embeddings $\rightarrow$ 2-D. DiRe preserves 3–4$\times$ more topological structure than UMAP (measured by Betti-curve DTW distance between reference and layout). UMAP wins on local-neighborhood metrics, consistent with the noise-memorisation account from Section 3.1. DiRe is 1.6$\times$ faster.

Topology error for a trial is $|b_1^{\text{fig-8}} - 2| + |b_1^{\text{torus}} - 2|$.

Pareto search. optuna 4.x NSGA-II sampler, population 25, seed 0. 150–200 trials per dataset. Search space in Section 2. Each trial fits DiRe on the target dataset, computes kNN accuracy with `KNeighborsClassifier`(15) on a fixed 70/30 train/test split, and computes topology error on the stress manifolds. Missing-value rows in OpenML datasets are dropped.

arXiv corpus. 723 457 arXiv papers embedded with BAAI/bge-small-en-v1.5 [13], chunk-level embeddings mean-pooled per paper to a 384-D vector. Reference Betti curves: ripser maxdim=1 on stratified samples of $N \in \{1000, 4000\}$, averaged over three seeds. Embedding Betti curves: ripser on the 2-D layout at the same stratified sample indices. DTW distance: `fastdtw` on the Betti curve time series.

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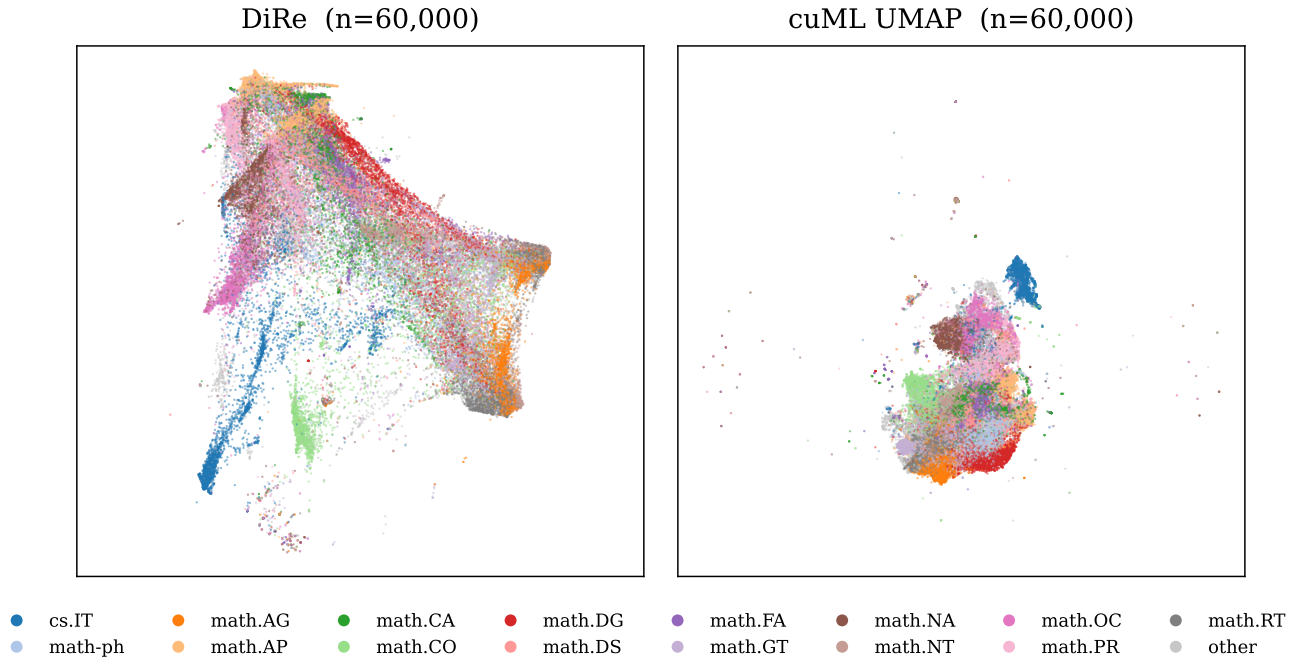

Figure 3: arXiv-paper 2-D layout, DiRe (left) vs cuML UMAP (right), coloured by primary arXiv category. Both methods recover category structure; UMAP produces tighter but more fragmented islands with visible gaps in long-range category similarity (e.g. math.DG vs math.SG). DiRe's layout is smoother and preserves the inter-category topology of the reference 384-D point cloud more faithfully.

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